

CLASSIFICATION SCHEME FOR AIRPLANE TARGETS BASED ON EMPIRICAL MODE DECOMPOSITION

Wang Baoshuai*, Du Lan*, Li Yanbing, Ruan Long, Liu Hongwei

*National Lab of Radar Signal Processing, Xidian University, Xi'an 710071, China

Abstract

This paper proposes a new method for the classification problem of turbojet aircraft, prop aircraft and helicopter under the condition of low-resolution radar system. Based on the jet engine modulation (JEM) characteristics of the aircrafts' radar echoes, we use the Empirical Mode Decomposition (EMD) algorithm to extract the three-dimensional micro-Doppler feature vector. Experimental results based on the simulated training data and the measured test data are presented, including the performance analysis for different Signal-Noise-Ratio (SNR) cases and comparison with some related methods.

Keywords: Micro-Doppler, Empirical Mode Decomposition (EMD), Jet engine modulation (JEM), Feature extraction.

1 Introduction

Since most conventional air defense radar systems belong to low-resolution radars whose range resolutions are much larger than the airplanes targets' sizes, an aircraft is usually regarded as a "point target". Thus it is difficult to categorize targets due to the limited information contained in the low-resolution echoes. However, compared with the traditional method in which the low-resolution radar is used to detect targets and the high-resolution radar is used to recognize targets, if we can make an initial target classification by using the low-resolution radar, we can not only reduce the complexity induced by the high-resolution radar but also make full use of the radar system available now. Thus it is an important research subject to find practical classification methods for low-resolution radar.

When radar transmits an electromagnetic signal to a moving target, the frequency of the received signal will be shifted from the frequency of the transmitted signal, known as the Doppler effect [1, 2]. If the target or any structural component of the target has an oscillatory or rotation motion in addition to the bulk motion of the target, the oscillatory or rotation motion will induce additional frequency modulation on the returned signal and generate side bands about the Doppler shifted frequency of the transmitted signal. The additional Doppler modulation is called the micro-Doppler effect [3, 4]. Different kinds of targets may have different kinds of micro-motions, which makes micro-motion, depicted by micro-Doppler modulation, a unique feature for different kinds of radar targets. Thus it is possible to realize target classification

by using the micro-motion features. The jet engine modulation (JEM) [5] characteristic, i.e., the periodical modulated echo produced by the high-speed rotating structures (rotating wings, fans or props) in an aircraft, is a kind of micro-Doppler effects for airplane targets which can be used to categorize airplane targets.

This paper proposes a novel feature extraction method based on Empirical Mode Decomposition (EMD) algorithm, through which we can separate the fuselage component and the JEM component in order to extract the discriminative features using the information within or between them. Experimental results show that our proposed method can achieve good classification performance under the test condition of relatively high signal-to-noise ratio (SNR).

2 Theoretical model of rotating structures in an aircraft

Assume the radar transmitted signal is $u(t) = \exp(j\omega_0 t)$ where ω_0 is the carrier frequency. Then the returned signal from the target can be expressed by $x(t) = a(t) \exp(j\omega_0 t + \phi(t)) = x_{\text{JEM}}(t) \exp(j\omega_0 t)$, where $a(t)$ is the amplitude function, $\phi(t)$ is the phase function and $x_{\text{JEM}}(t)$ is the complex envelope which includes the information of amplitude modulation and phase modulation. Then according to [5], we can get the ideal JEM Doppler spectrum as follows:

$$X_{\text{JEM}}(f) = \sum_{m=1}^N \sum_{i=-N_{1m}}^{N_{1m}} c_{m,i} \delta(f - f_d + i q_m n_m f_{rm}) \quad (1)$$

Where N = number of rotating structures.

$\delta(\cdot)$ = unit impulse function,

f_{rm} = rotation cyclic frequency of the $(m+1)th$ rotating structure,

$q_m = 1$ for even blades, 2 for odd blades.

$c_{m,i}$ = amplitude of each spectrum line.

Equation (1) shows that the ideal JEM Doppler spectra consist of N_{1m} pairs of spectrum lines about the centre frequency of the target. The spectrum lines are separated by $f_{rm} = q_m n_m f_{rm} \cdot N_{1m}$ and B_{1m} denote the number of spectrum lines and the bandwidth at the single side respectively[5]:

$$N_{1m} = \frac{8\pi(L_{2m} - L_{1m}) \cos \beta'}{q_m n_m \lambda} \quad (2)$$

$$B_{1m} = \frac{8\pi f_{rm} (L_{2m} - L_{1m}) \cos \beta'}{\lambda} \quad (3)$$

Where L_{1m} and L_{2m} respectively denote the distance of the blade roots and that of the tips from the centre of rotation in the $(m+1)$ th rotating structure, β represent the pitching angle between the plane of rotation and the line-of-sight (LOS) from the radar to the centre of rotation. When the rotating plane is parallel to the airplane's flying direction, $\beta' = \beta$; when they are vertical $\beta' = \pi/2 - \beta$.

According to the airplane aerodynamics, the linear velocities of the blade tips of the three kinds of aircrafts are almost the same. Since the lengths of the blades are quite different for the three kinds of aircrafts, according to equation (2) and (3), we can find some differences in the JEM Doppler spectra of the three kinds of aircrafts, i.e. the Doppler spectra of a turbojet aircraft has least spectrum lines and the spectrum lines are sparsest, the Doppler spectra of a helicopter has most spectrum lines and the spectrum lines are densest and the spectra properties of a prop aircraft are in the compromising position between those of a turbojet aircraft and a helicopter.

3 Feature extraction based on Empirical Mode Decomposition algorithm

The Empirical Mode Decomposition (EMD) method was first pioneered by Huang et al. to adaptively decompose nonlinear and non-stationary time series in 1998[6]. Being quite different from other signal decomposition techniques, e.g., Fourier transform and wavelet transform, which map the signal space onto a space spanned by a set of predefined basis, the general idea of EMD is to adaptively decompose a signal into a number of "basic function" termed intrinsic mode functions (IMFs), which are derived directly from the signal. According to the "sifting process" defined in [6], for a given signal x , EMD ends up with a representation of the form:

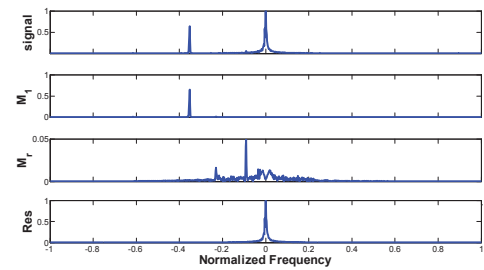
$$x = \sum_{i=1}^L \text{imf}_i + r_L \quad (4)$$

where imf_i denotes the i th IMF component and r_L is the residual after L iterations.

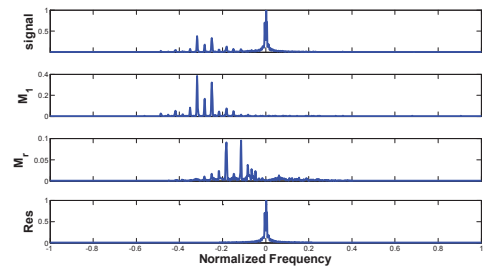
In its original formulation, the EMD can only be applied to real-valued time series, which is not available for several important engineering areas (sonar, radar, to mention but a few) that use complex-valued signal. In [7], Rilling et al. developed an extension of EMD suitable for dealing with complex-valued data. The main idea of this extension is that it considers the real-valued signal as the scale of its local oscillations, which means fast oscillations superimposed on slow oscillations, and the complex-valued signal as the scale of its local rotation, which means fast rotations superimpose on slow rotations. Flandrin et al. [8] point out that EMD acts essentially as a filter bank resembling those involved in wavelet decomposition but it is worth stressing that this filter bank corresponds to an adaptive time-variant filter than a pre-determined one.

Fig. 1 shows the Doppler spectra of the original signal, IMFs and residual for the measured data of the turbojet aircraft,

prop aircraft and helicopter. In Fig. 1 (a), (b) and (c), the first row shows the Doppler spectra of the original signal, the second row is $M_1 = |\text{FFT}(\text{imf}_1)|$, where the function $\text{FFT}(\bullet)$ denotes the fast Fourier transformation, the third row depicts $M_r = |\text{FFT}(\sum_{i=2}^L \text{imf}_i)|$, and the Doppler spectra of the residual component is shown in the last row. From Fig. 1 (a) and (c), we can see that, for both turbojet aircraft and helicopter, after applying EMD to the aircraft echoes, the JEM component is effectively separated from the fuselage component. Since a turbojet aircraft flies much faster than a helicopter, when the PRF is relatively high, the fuselage component of the turbojet aircraft can be regarded as high-frequency component and that of helicopter as low-frequency component. After applying EMD to the returns of a turbojet aircraft, M_1 captures the high-frequency fuselage component, and M_r captures the JEM component. On the contrary, for a helicopter, M_1 contains the JEM component, while M_r mainly captures the low-frequency fuselage component. From Fig. 2 (b), we can see that for the prop aircraft, the EMD method cannot separate the fuselage component with JEM component effectively, i.e., M_1 contains the fuselage component and a part of the JEM component, and the rest of the JEM component is left in M_r . Additionally, from Fig. 1 we can see that the ground clutter is always captured by the residual component for the three kinds of aircrafts; therefore, we can achieve the clutter suppression by just discarding the residual component. It is another advantage of the EMD method in this application.



(a)



(b)

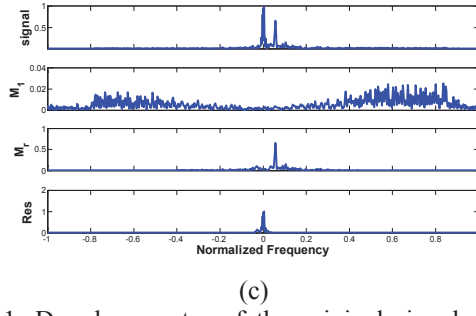


Figure 1. Doppler spectra of the original signal, IMFs and residual from the measured data: (a) Turbojet aircraft; (b) Prop aircraft; (c) Helicopter.

Based on the analysis of Fig. 1, here we extract the following three-dimensional features to distinct the differences in the component M_1 and M_r .

- Feature 1:

$$F(1) = \text{entropy}(M_1) = -\sum_{f=1}^T p_f \log(p_f) \text{ where } T \text{ is the length of the signal,}$$

$$p_f = \frac{M_1(f)}{\sum_{f=1}^T M_1(f)} \text{ and } M_1 = \{M_1(f)\}_{f=1}^T, \text{ which}$$

represents the entropy of M_1 . For a series, the entropy indicates the energy distribution, the more the energy gathers, the less the entropy is.

- Feature 2:

$$F(2) = \sum_{f=1}^T (l - \bar{f})^2 \bar{M}_r(f) \text{ with,}$$

$$\bar{f} = \sum_{f=1}^T f \times \bar{M}_r(f),$$

$$M_r = \{M_r(f)\}_{f=1}^T \text{ and } \bar{M}_r(f) = \frac{M_r(f)}{\sum_{f=1}^T M_r(f)} \text{ which}$$

represents the 2nd order central moment of M_r . The 2nd order central moment describes the distribution of a series relative to its centroid.

- Feature 3:

$$F(3) = \frac{\text{energy}(M_1)}{\text{energy}(M_r)} = \frac{\|M_1\|_2^2}{\|M_r\|_2^2}, \text{ which represents the energy ratio between } M_1 \text{ and } M_r.$$

Fig. 4 shows the two-dimensional feature distribution for the measured three kinds of aircrafts. We can see that the proposed features are well distinguished in the feature domain.

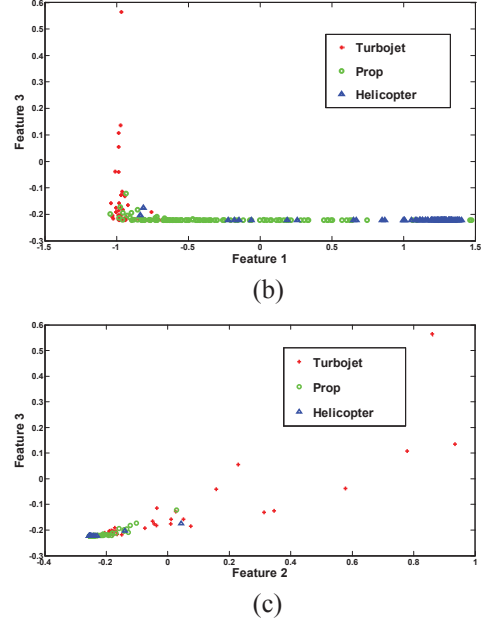
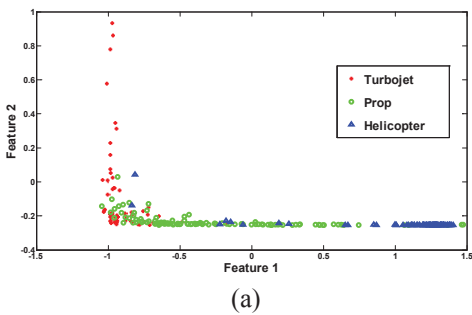


Figure 4: The two-dimensional feature distribution for the measured three kinds of aircrafts: (a) Feature 1 and Feature 2; (b) Feature 1 and Feature 3; (c): Feature 2 and Feature 3.

4 Experiment Results

In this experiment, we use the simulated data in the training stage and the measured data in the test stage to extract the three-dimensional features described above. The SVM with Gaussian kernel is chosen as the classifier and the parameter of Gaussian kernel is optimized to have the best classification result by searching in a certain range.

The data in the training stage is simulated based on the theoretical model described in section 2. Table 1 and Table 2 list the simulation parameters of the rotating structures of the 15 aircrafts (5 turbojet aircrafts, 5 prop aircrafts and 5 helicopters) and the scene parameters respectively. In addition, since the echoes of helicopters are mainly affected by the main rotor which is larger in size than the tail rotor. In our simulation, we simply set the length of the tail rotor is 1/3 of that of the main rotor and the corresponding rotation speed is 3 times that of the main rotor.

The measured data are collected from many airplane targets of all the three kinds, i.e. turbojet aircraft, prop aircraft and helicopter.

In the following experiment, we compare our classification method with the two related methods which are described as follows.

- Method 1: This method is proposed in the recent paper [9], which uses the eigenvalue spectrum of the aircraft target echo to extract the four-dimensional JEM feature vector, i.e., the number of larger eigenvalues that contain 98% of the eigenvalue spectrum's energy, the modified periodic factor, the entropy of the eigenvalue spectrum and the sum of eigenvalues normalized by the largest eigenvalue.
- Method 2: This method is proposed in [10]. We select three features (i.e., the entropy of the Doppler spectra of

radar echoes, the 2nd order central moment of the Doppler spectra of radar echoes and the 4th order central moment of the Doppler spectra of radar echoes) which are often used in the low-resolution radar target classification area from the several features [10] proposed.

Additionally, we use the moving-target indicator (MTI) filter with a double canceler [11] to suppress the ground clutter before feature extraction for the two methods mentioned above.

Aircraft index	Rotate speed(r/min)	L1(m)	L2(m)	Blade number
T - 1	3520	0.38	1.10	38
T - 2	8615	0.18	0.51	27
T - 3	3000	0.30	1.00	30
T - 4	5000	0.20	0.60	33
T - 5	4000	0.24	0.80	42
P - 1	1200	0.68	1.70	4
P - 2	1245	0.79	0.95	4
P - 3	2340	0.23	0.66	3
P - 4	3400	0.27	0.76	3
P - 5	6060	0.12	0.33	2
H - 1	123	0	16.0	8
H - 2	242	0	8.60	5
H - 3	192	0	10.65	5
H - 4	217	0	7.80	4
H - 5	394	0	5.35	3

Table 1: Rotation speeds and configuration parameters of the rotating structures of 5 turbojet aircrafts (T), 5 prop aircrafts (P) and 5 helicopters (H)

	Turbojet	Prop	Helicopter
Range(m)	30000 - 40000	30000 - 40000	30000 - 40000
Altitude(m)	15000 - 25000	4000 - 12000	500 - 10000
Speed(m/s)	150 - 200	100 - 150	50 - 100

Table 2: Typical scene parameters

	Proposed method	Method 1	Method 2
Turbojet	84.75%	84.75%	62.71%
Prop	72.90%	56.07%	63.55%
Helicopter	92.77%	89.16%	100.00%
Average	82.33%	73.89%	75.50%

Table3: Classification accuracies of each aircraft category and the average rates via the three methods.

Table3 compares the classification accuracies of each aircraft category and the average rates via the proposed method, method 1 and method 2. Since the measured data are collected under high SNR condition, the noise in the classification stage can be ignored. From Table 3 we can see that, Method 1 cannot categorize the prop aircraft effectively (its classification accuracy is 56.07%) and Method 2 fails to categorize both the turbojet aircraft and the prop aircraft effectively (the classification accuracy for turbojet aircraft is 62.71% and that for prop aircraft is 63.55%), while our proposed method can categorize any aircraft of the three kinds into the right class with the accuracy over 70%.

In practice, the robustness to noise is an important evaluation for a classification algorithm. In order to evaluate the

performance of the proposed method under different SNR conditions, here we add zero-mean Gaussian white noise to the measured data. Fig. 5 shows the curves of the classification rates varying with SNR. We note that under the SNR condition of 5dB or 10dB, the proposed method cannot perform as well as Method, however, when the SNR is above 15 dB, the proposed method achieve best classification performance.

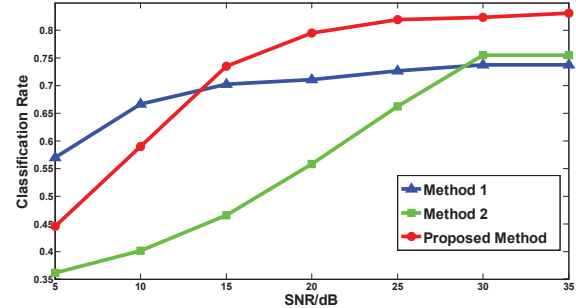


Figure 5: Variation of the classification rates of the measured test data with the test SNR via the proposed method, Method 1 and Method 2.

5 Conclusion

This paper studies on the classification problem of turbojet aircraft, prop aircraft and helicopter under the condition of low-resolution radar system. Firstly, we find the differences in the Doppler spectra of the three kinds of aircrafts based on the theoretical model of the rotating structures in the aircrafts. Then the EMD algorithm is used to separate the fuselage component with the JEM component, thereby we extract the discriminative features from the EMD results. Finally, experimental results with the simulated training data and the measured test data are proposed, which shows that our proposed method can achieve good classification performance under the test condition of relatively high signal-to-noise ratio (SNR).

Acknowledgements

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